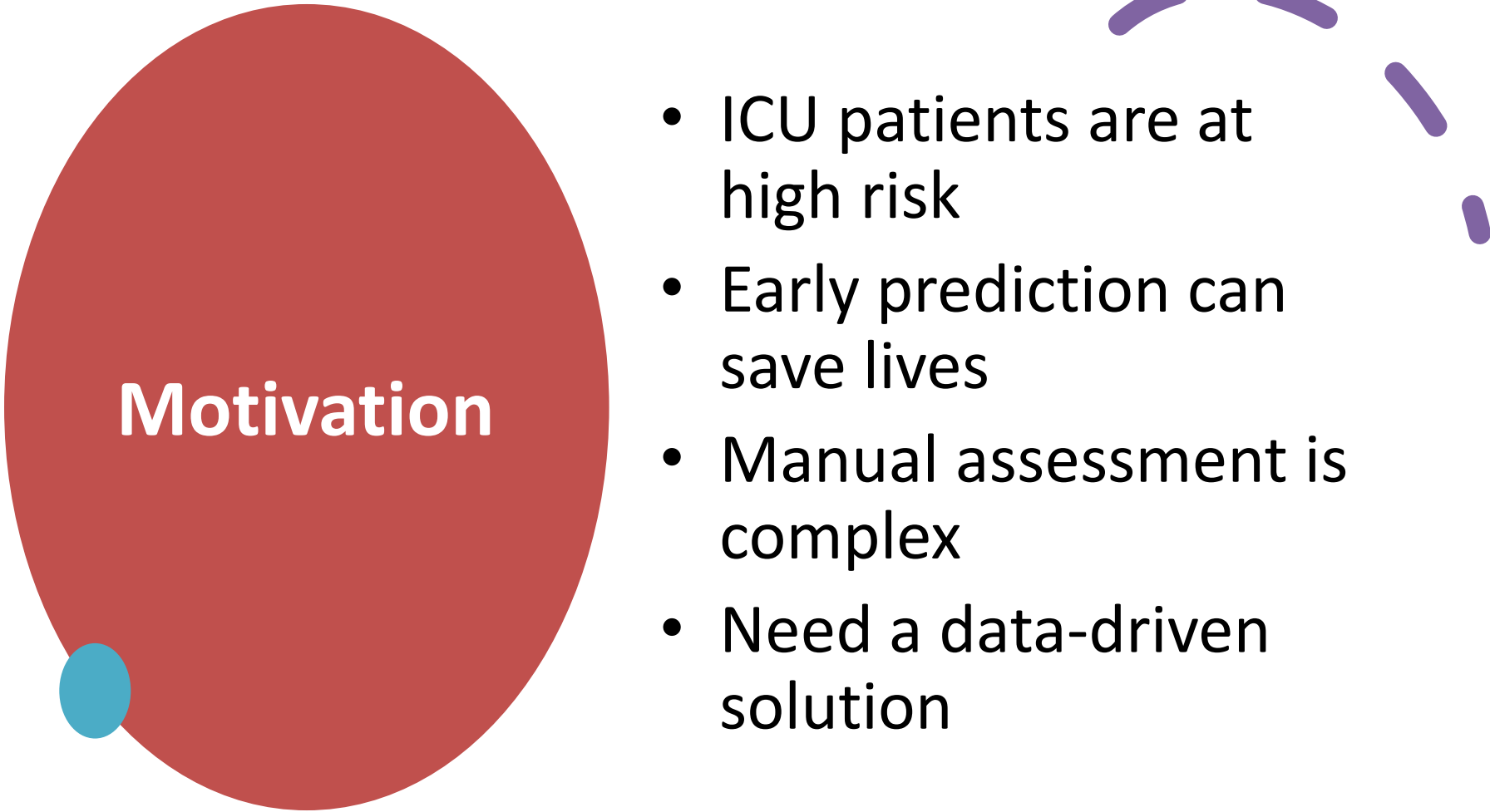


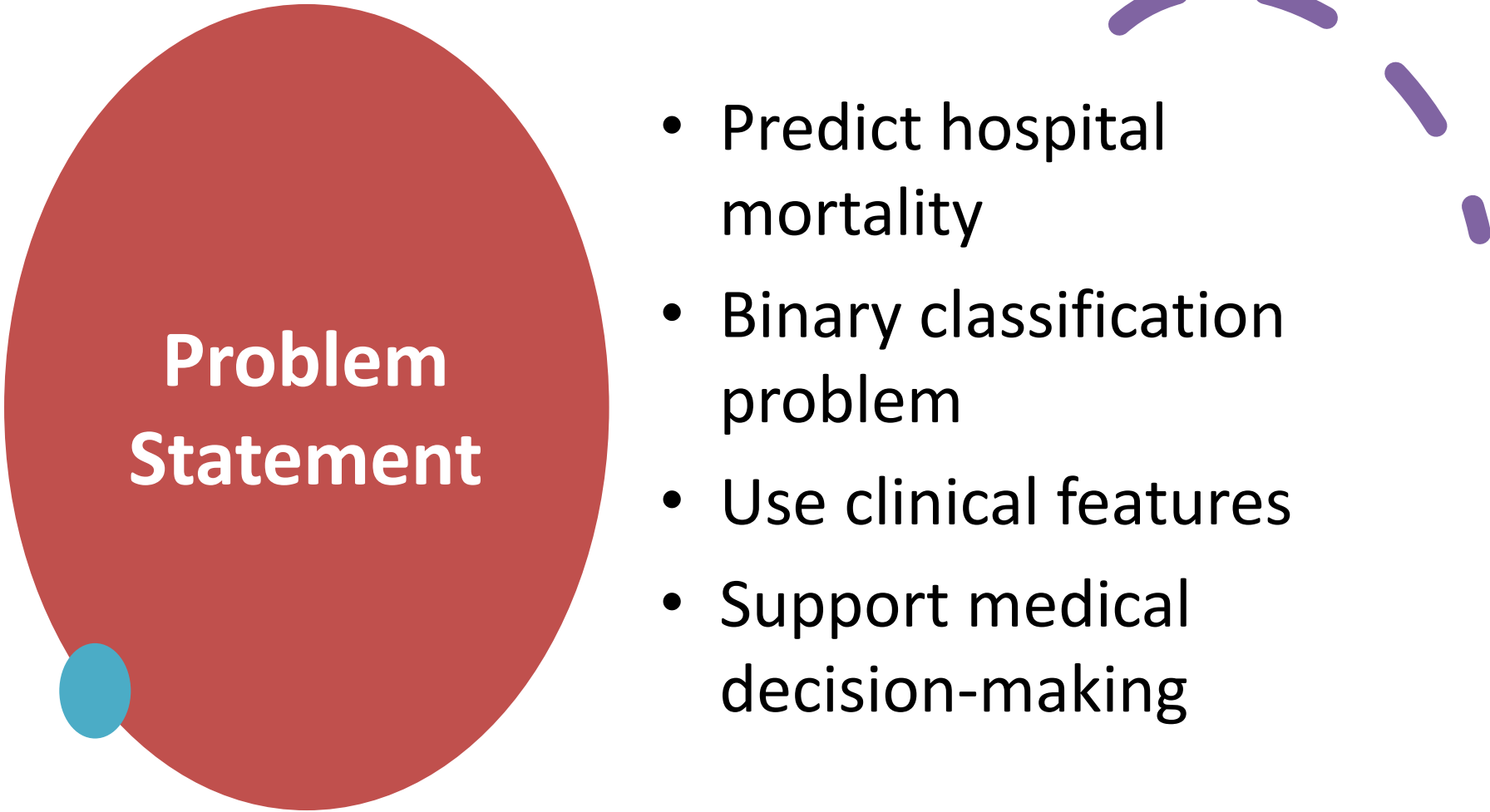
ICU Mortality Prediction using Machine Learning

Siva Ram Potluri
UMBC Data Science
Capstone
Instructor:
Dr. Chaojie (Jay)
Wang



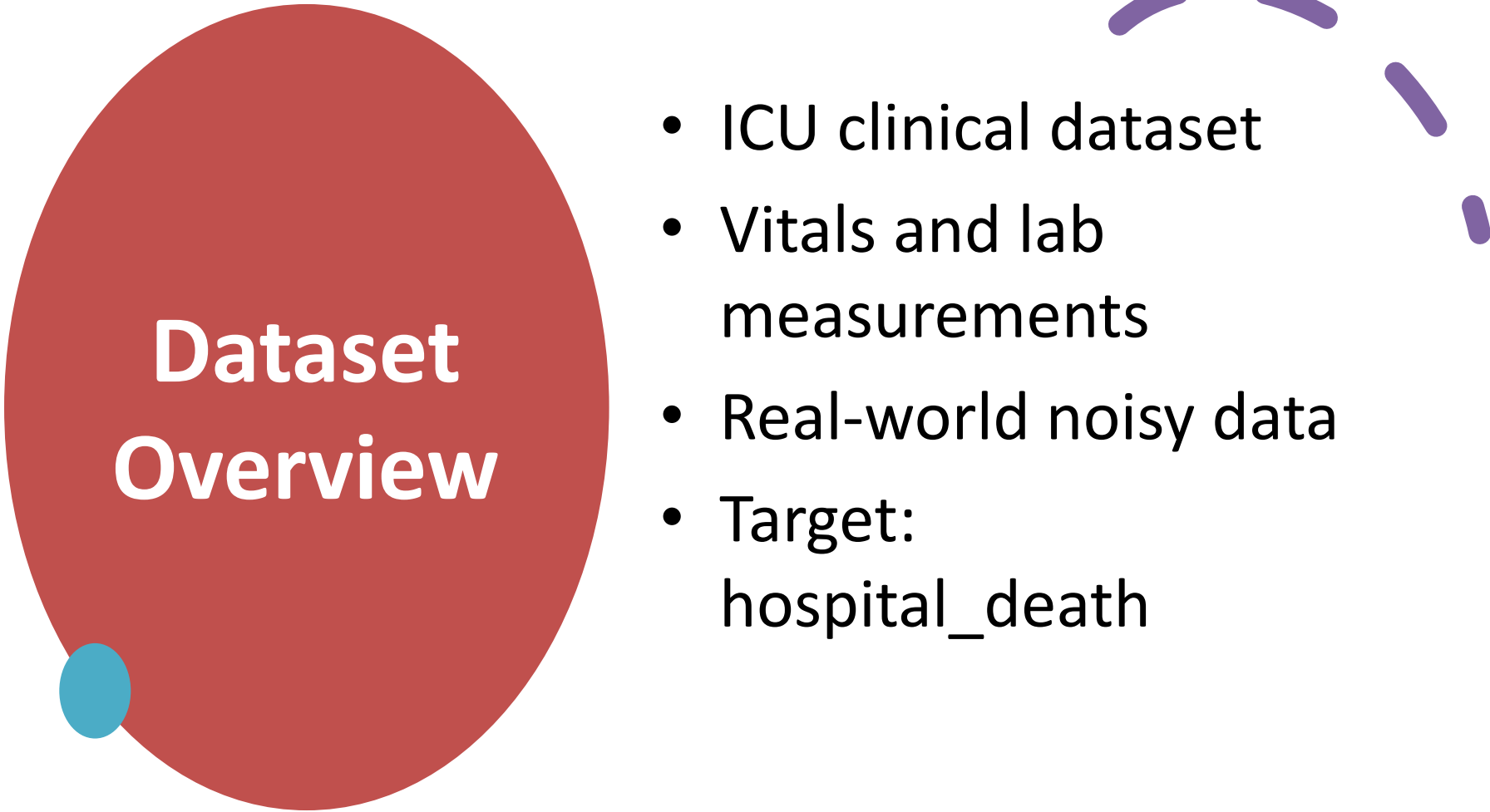
Motivation

- ICU patients are at high risk
- Early prediction can save lives
- Manual assessment is complex
- Need a data-driven solution



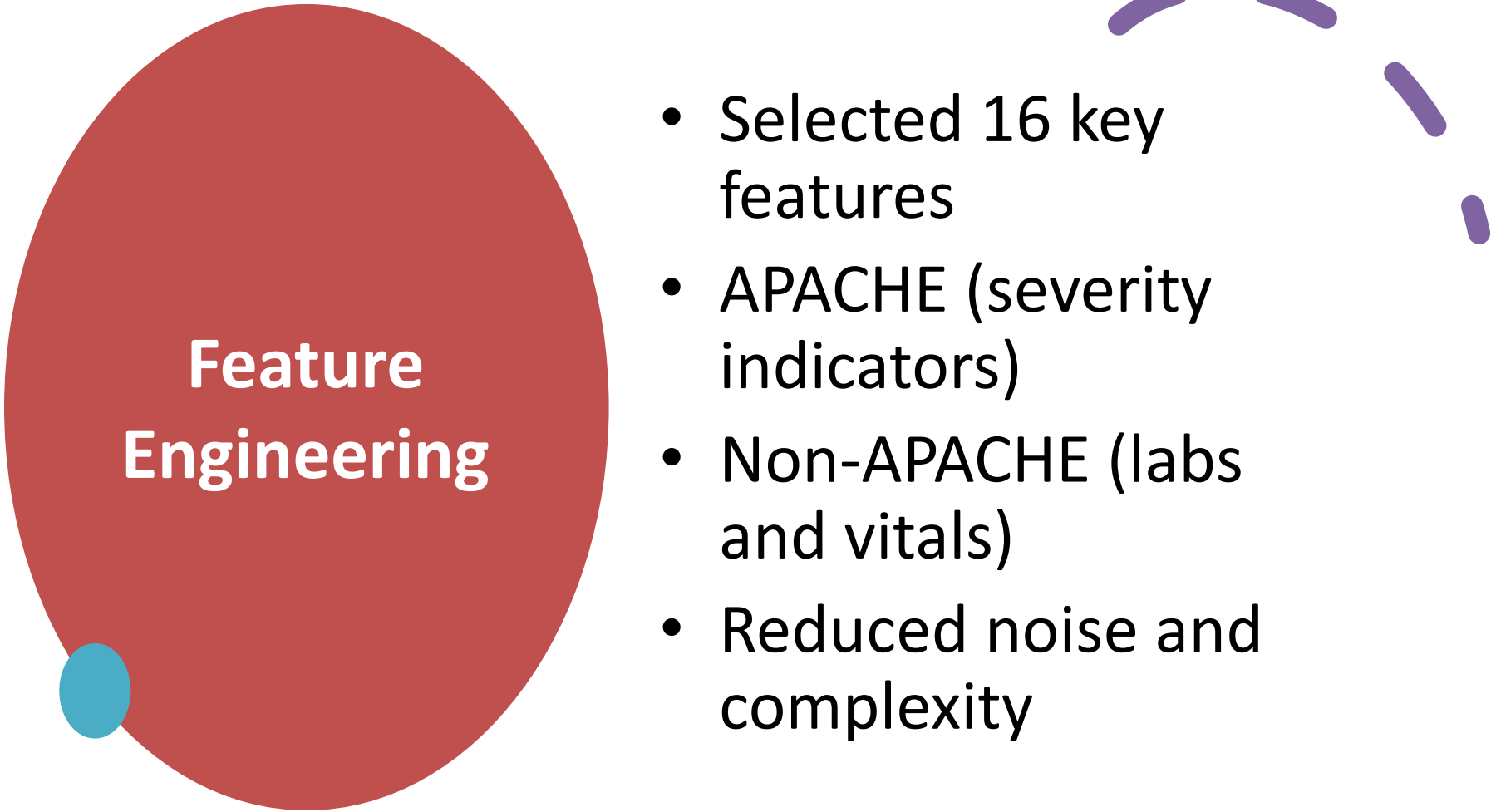
Problem Statement

- Predict hospital mortality
- Binary classification problem
- Use clinical features
- Support medical decision-making



Dataset Overview

- ICU clinical dataset
- Vitals and lab measurements
- Real-world noisy data
- Target:
hospital_death



Feature Engineering

- Selected 16 key features
- APACHE (severity indicators)
- Non-APACHE (labs and vitals)
- Reduced noise and complexity



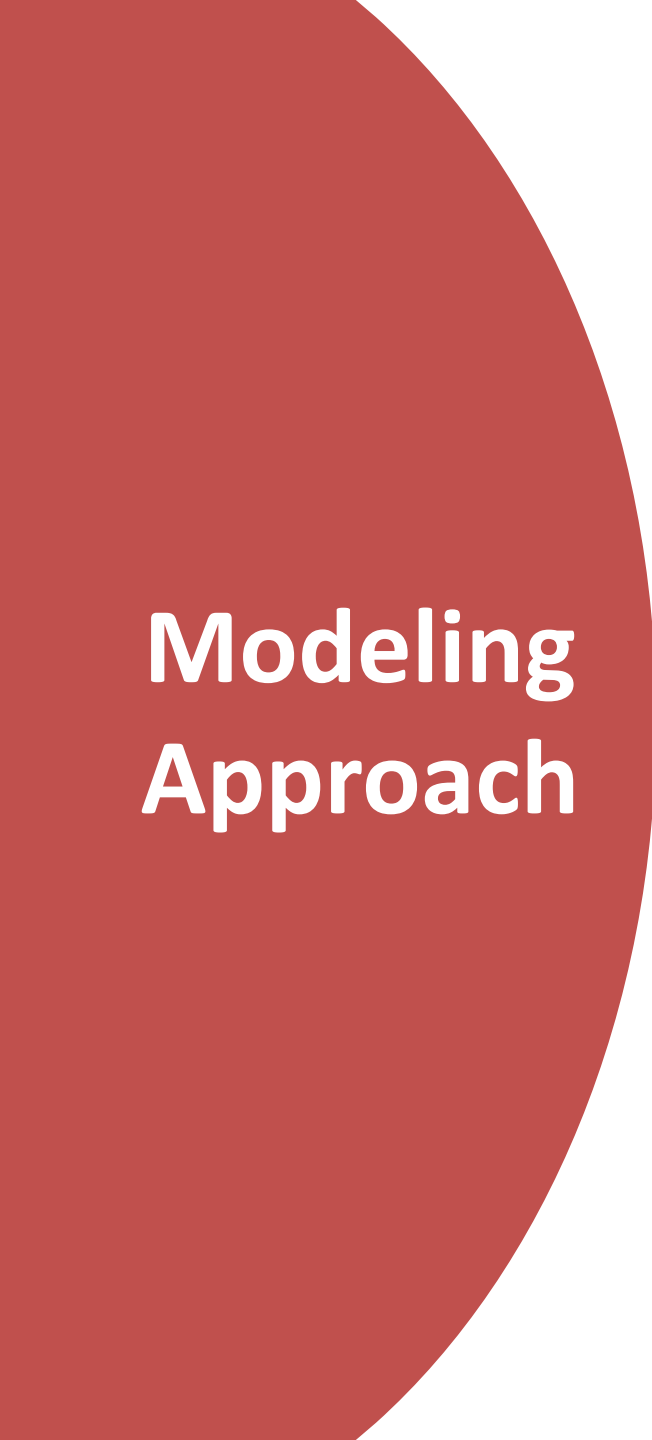
Exploratory Data Analysis

- 
- Distribution analysis
 - Boxplots for comparison
 - Correlation heatmap
 - Reveals data patterns



Key Insights

- Class imbalance observed
 - Some features separate outcomes
 - Presence of outliers
 - Moderate correlations
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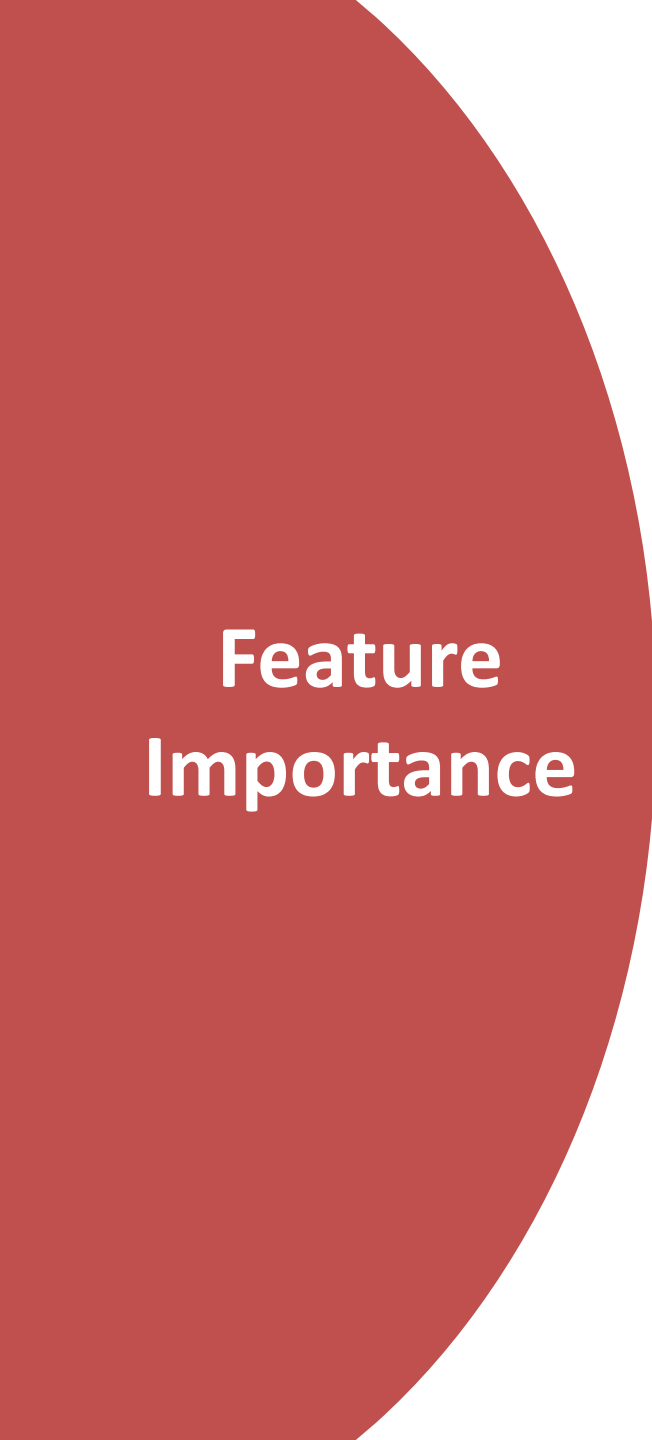
Modeling Approach

- Logistic Regression
 - Random Forest
 - Gradient Boosting
 - Compared using ROC-AUC
- 
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Results

- ROC-AUC comparison
- Gradient Boosting performed best
- Handles complex patterns
- Best predictive performance

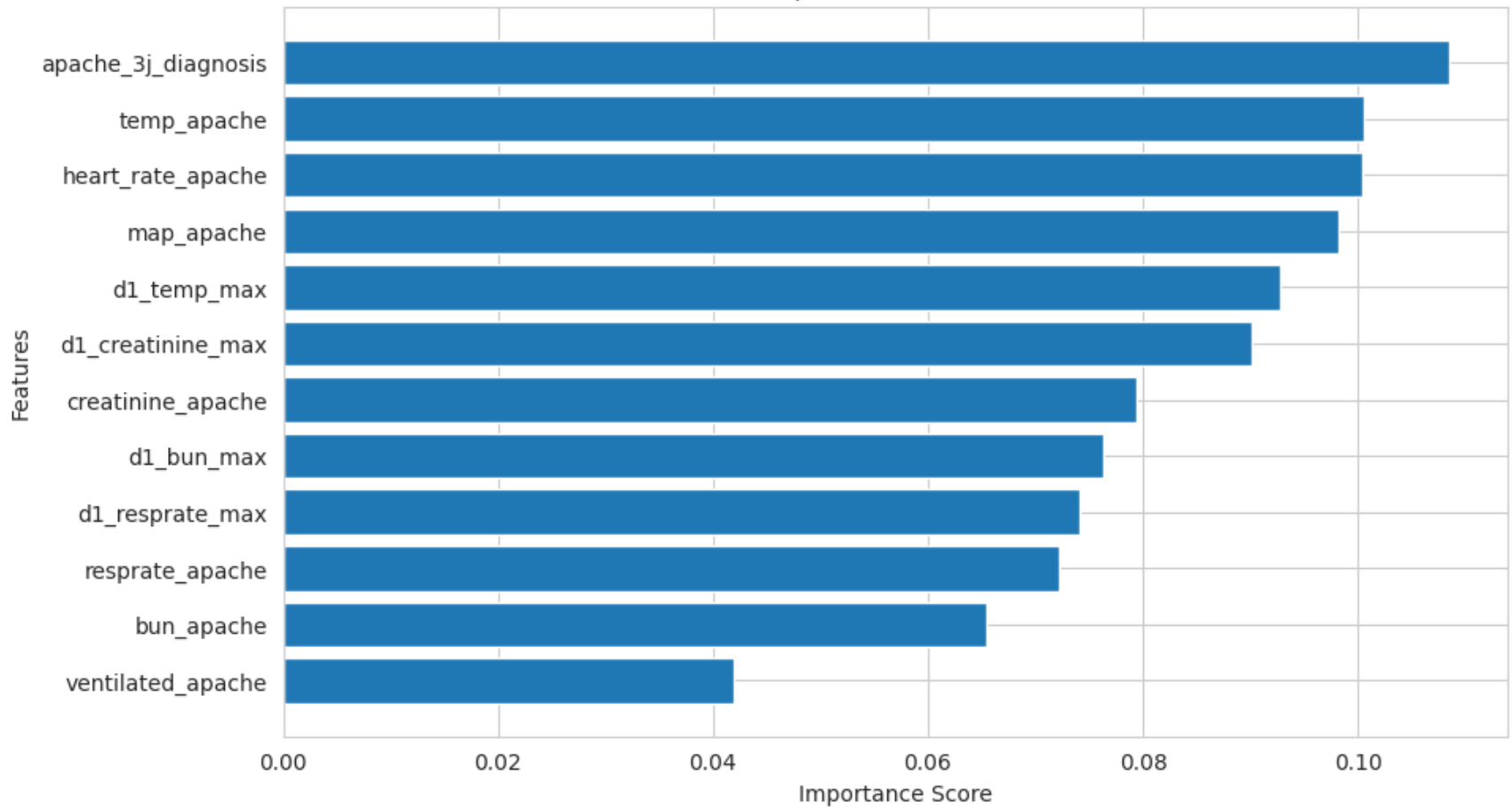




Feature Importance

- Top features identified
 - APACHE features most important
 - Improves interpretability
 - Supports decision-making
- 

Feature Importance (Random Forest)



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Interactive Dashboard

- Built using Streamlit
 - Real-time predictions
 - Visual outputs (charts, gauge)
 - User-friendly interface
- 
- Four short, thick purple lines of varying lengths and orientations arranged in a curved pattern in the bottom right corner.



ICU Mortality Prediction Dashboard

Predict hospital mortality risk using selected **APACHE** and **non-APACHE** clinical features.

Dataset Rows

91,713

Features Used

12

Mortality Rate

8.6%

Visualiza

Plotly

Project Overview

This dashboard presents the full ICU mortality prediction workflow:

- dataset exploration
- feature interpretation
- model comparison
- real-time mortality risk prediction

The objective is to identify patients at higher risk using structured clinical variables.

APACHE Features

8 variables

Non-APACHE Features

4 variables

Target Distribution

Hospital Outcome Distribution



Conclusion

- Model predicts ICU mortality effectively
 - Gradient Boosting is best model
 - Useful for healthcare support
 - Provides actionable insights
- 



Future Scope

- Use larger datasets
 - Apply deep learning models
 - Integrate with hospital systems
 - Improve feature engineering
- 

Thank you
